DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Applied IOT

CIA 3

Smart Street Lighting and Energy Optimization

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INTRODUCTION

Street lighting plays an essential role in ensuring safety and visibility during nighttime in urban and rural environments. However, traditional street lighting systems operate continuously regardless of environmental conditions or road activity, which results in unnecessary energy consumption and increased operational costs. With the growing demand for energy efficiency and sustainable infrastructure, there is a need for intelligent systems that can optimize the operation of street lights.

The integration of Internet of Things (IoT) technologies enables the development of smart systems capable of monitoring environmental conditions and automatically controlling devices. By combining sensors, microcontrollers, and cloud platforms, it is possible to create intelligent street lighting systems that operate only when necessary.

This project proposes a Smart Street Lighting and Energy Optimization System that automatically controls street lights based on ambient light and motion detection. An LDR sensor is used to detect environmental light conditions, while a PIR motion sensor detects the presence of pedestrians or vehicles. A current sensor monitors energy consumption, and the ESP32 microcontroller processes sensor data and controls the lighting system. The system also transmits real-time data to the ThingSpeak cloud platform, enabling remote monitoring and analysis of energy usage.

PROBLEM STATEMENT

Conventional street lighting systems operate on fixed schedules and remain active throughout the night regardless of whether roads are being used. This results in unnecessary energy consumption, increased electricity costs, and inefficient energy management.

In many locations, street lights remain turned on even when there are no pedestrians or vehicles nearby. Additionally, the lack of real-time monitoring prevents authorities from analyzing power consumption patterns or detecting faults in the system.

Therefore, there is a need for an intelligent street lighting system capable of automatically controlling lights based on environmental conditions and movement detection. Such a system should also provide real-time monitoring and energy usage analysis through cloud platforms.

OBJECTIVES

The main objectives of this project are:

- To design and implement an IoT-based smart street lighting system.
- To automatically control street lights using ambient light detection and motion sensing.
- To monitor energy consumption using a current sensor.
- To provide local monitoring using an OLED display.
- To transmit system data to the ThingSpeak cloud platform for remote monitoring.
- To demonstrate energy optimization through automated street lighting control.

SYSTEM OVERVIEW

The proposed system integrates sensing, processing, display, and cloud communication components to form an intelligent street lighting solution.

The LDR sensor detects ambient light levels and determines whether it is day or night. When the environment becomes dark, the system activates motion detection using the PIR sensor.

If motion is detected, the ESP32 microcontroller turns on the LED street light. A current sensor measures the energy consumed by the LED, and the data is displayed locally on the OLED display.

At the same time, the ESP32 sends the collected sensor data to the ThingSpeak cloud platform, where it is stored and visualized in graphical form. This enables remote monitoring and analysis of system performance and energy consumption.

WORKING PRINCIPLE

The working principle of the system is based on automated sensing and IoT communication.

Initially, the system reads the ambient light level using the LDR sensor. If the light intensity is above the predefined threshold, the system identifies it as daytime and keeps the street light turned off.

When the light intensity falls below the threshold, indicating nighttime conditions, the system activates the PIR motion sensor. If the PIR sensor detects movement from pedestrians or vehicles, the ESP32 microcontroller turns on the LED street light.

Once the light is activated, the current sensor measures the power consumption of the LED. The measured values are displayed on the OLED display and simultaneously transmitted to the ThingSpeak cloud platform for remote monitoring.

This process repeats continuously, ensuring real-time monitoring and efficient energy usage.

CLOUD INTEGRATION

Cloud integration plays an important role in enabling remote monitoring and data visualization for the smart street lighting system.

The ESP32 microcontroller connects to the internet through Wi-Fi and sends sensor data to the ThingSpeak cloud platform using an API-based communication method. The transmitted data includes LDR values, motion detection status, current consumption, and LED status.

Once the data is uploaded to ThingSpeak, it is stored and presented in graphical format. These graphs allow users to analyze energy consumption patterns and monitor system performance remotely.

Cloud connectivity enhances the functionality of the system by providing centralized monitoring, historical data analysis, and improved energy management.

HARDWARE / COMPONENT SELECTION

1. **ESP32 Microcontroller :**

   The ESP32 serves as the central controller of the system. It processes sensor data and provides Wi-Fi connectivity for IoT communication.

2. **LDR Sensor :**

   The Light Dependent Resistor detects ambient light levels and determines whether it is day or night.

3. **PIR Motion Sensor :**

   The PIR sensor detects human or vehicle movement near the street

   light.

4. **Current Sensor :**

   The current sensor measures the power consumption of the LED street

   light.

5. **OLED Display :**

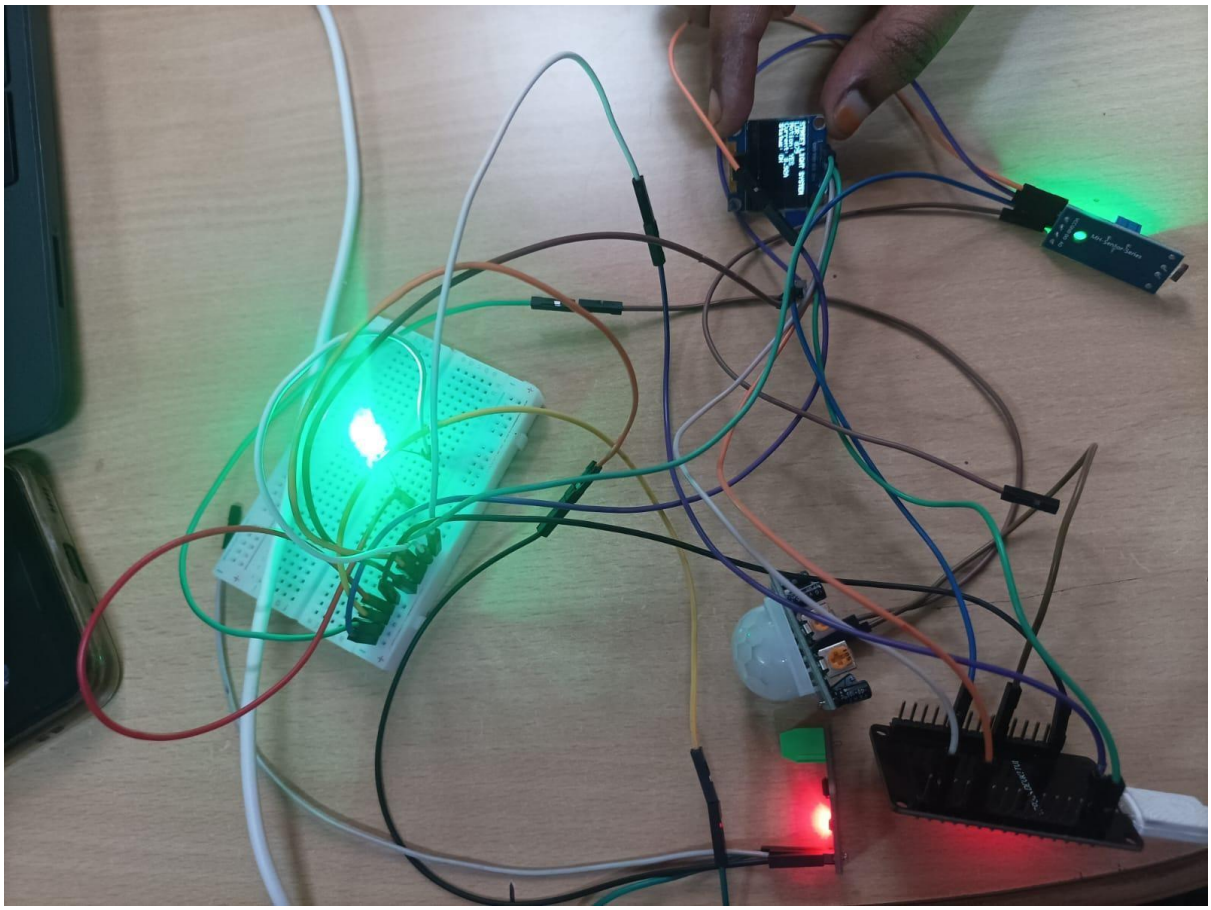
   The OLED display shows real-time system information such as sensor readings, LED status, and energy usage.

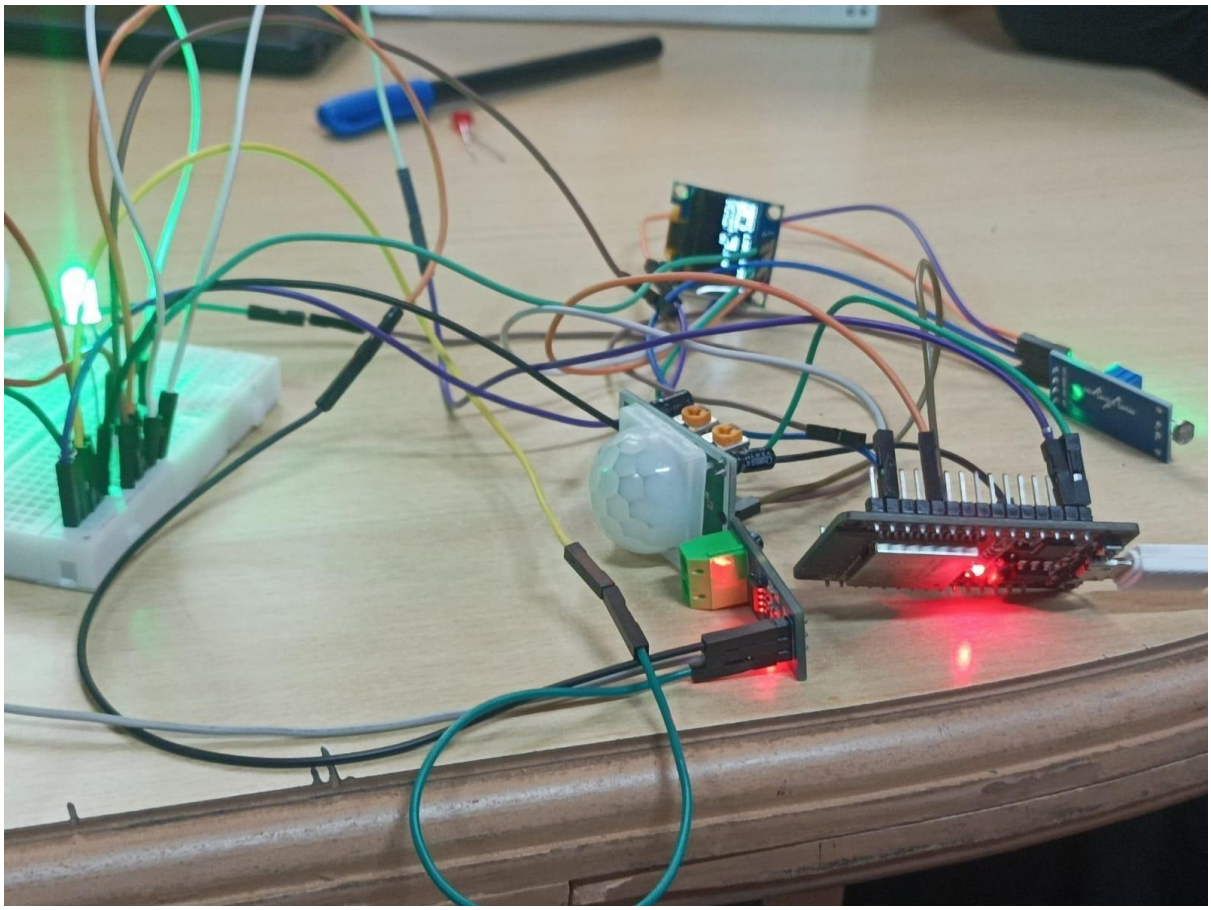
6. **LED :**

   The LED represents the street light in the prototype system.

IMPLEMENTATION

- The system was implemented using an ESP32 microcontroller interfaced with an LDR sensor, PIR motion sensor, current sensor, OLED display, and LED.
- The sensors were connected to the ESP32 through appropriate GPIO pins. The LDR sensor measures ambient light intensity, while the PIR sensor detects motion. The current sensor measures the electrical current consumed by the LED.
- The ESP32 was programmed using the Arduino IDE to continuously read sensor data and control the LED street light accordingly. The OLED display shows real-time system information such as sensor readings and LED status.
- The ESP32 also establishes a Wi-Fi connection and periodically uploads sensor data to the ThingSpeak cloud platform for remote monitoring.





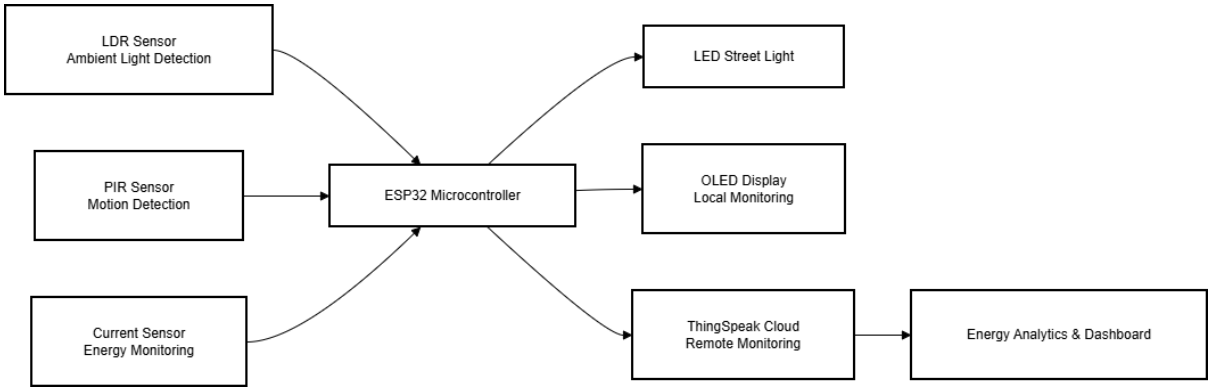
SOURCE CODE :

```
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <WiFi.h>
#include <ThingSpeak.h>


#define LDR_PIN 34
#define PIR_PIN 27
#define CURRENT_PIN 35
#define LED_PIN 2   // PWM pin
#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64


const char* ssid = "Oneplus";
const char* password = "Magic9886@";
unsigned long channelID = 3280844;  // ThingSpeak Channel ID
const char* writeAPIKey = "0SQ7N0POPA014GCR";


Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire, -1);
```

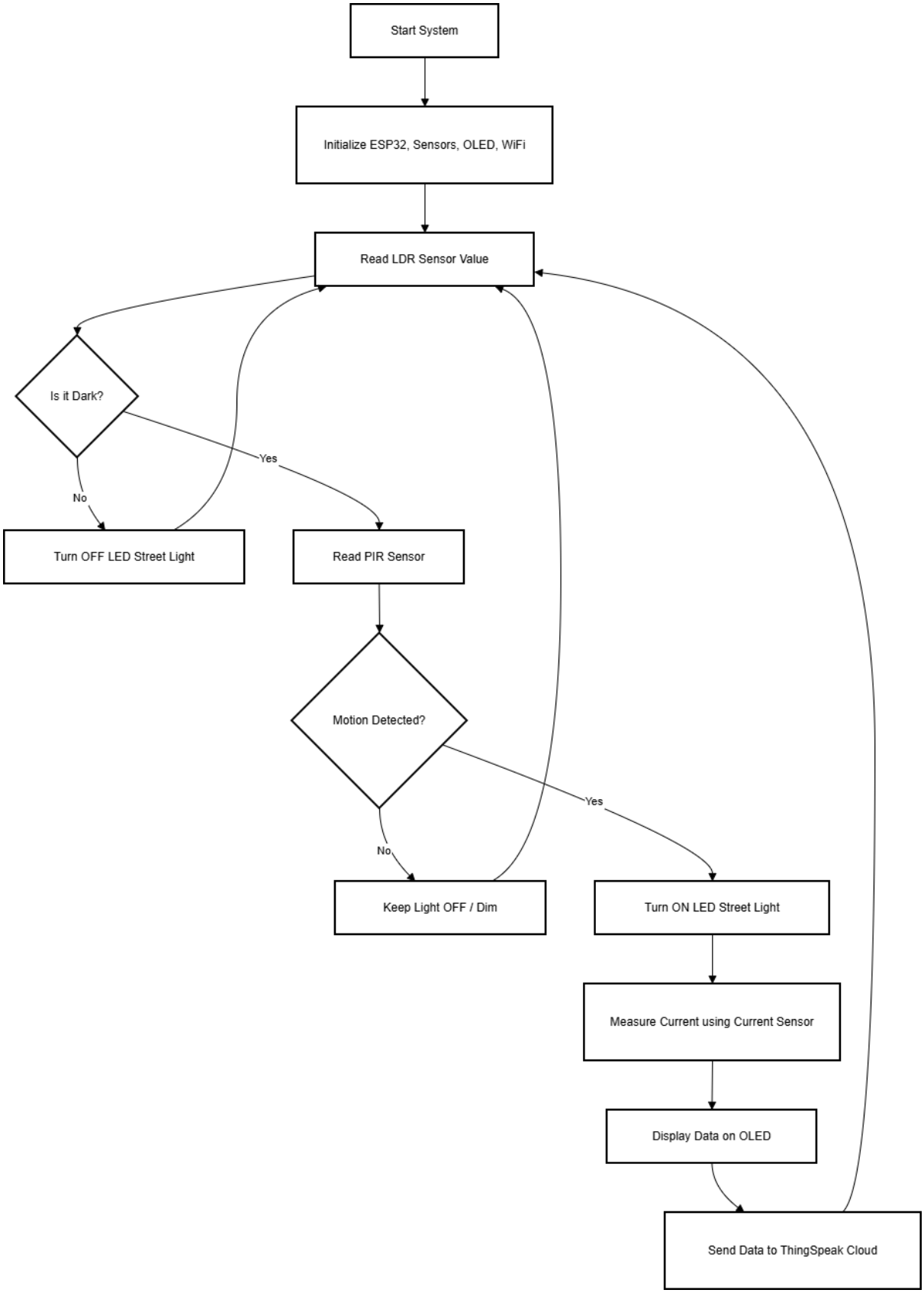
```
const int DARK_THRESHOLD = 1500;


int ledChannel = 0;
int ledFreq = 5000;
int ledResolution = 8;

WiFiClient client;

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);


  ledcAttach(LED_PIN, ledFreq, ledResolution);


  if(!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("OLED failed"));
    for(;;);
  }
  display.clearDisplay();
  display.setTextColor(WHITE);


  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");
  while(WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nConnected to WiFi");

  ThingSpeak.begin(client); // Initialize ThingSpeak
}

void loop() {
  int ldrValue = analogRead(LDR_PIN);
  int motion = digitalRead(PIR_PIN);
  float current = getACSCurrent(CURRENT_PIN);
  int brightness = 0;


  if (ldrValue < DARK_THRESHOLD) {
    if (motion == HIGH) {
      brightness = 255; // Full brightness if motion detected
    } else {
      brightness = 50;  // Dim mode
    }
  } else {
    brightness = 0;      // Day time, light OFF
  }

  ledcWrite(ledChannel, brightness);
  updateOLED(ldrValue, motion, current, brightness);
```

```
  ThingSpeak.setField(1, ldrValue);
  ThingSpeak.setField(2, motion);
  ThingSpeak.setField(3, current);
  ThingSpeak.setField(4, brightness);
  int response = ThingSpeak.writeFields(channelID, writeAPIKey);
  if(response == 200){
    Serial.println("ThingSpeak update successful");
  } else {
    Serial.println("ThingSpeak update failed");
  }

  delay(15000); // ThingSpeak allows 15s minimum update interval
}
float getACSCurrent(int pin) {
  int raw = analogRead(pin);
  float voltage = (raw / 4095.0) * 3.3; // ESP32 12-bit ADC
  return (voltage - 1.65) / 0.185;      // ACS712-05B sensitivity
}
void updateOLED(int ldr, int mot, float curr, int br) {
  display.clearDisplay();
  display.setCursor(0,0);
  display.println("STREET LIGHT SYSTEM");
  display.print("LDR: "); display.println(ldr);
  display.print("Motion: "); display.println(mot ? "YES" : "NO");
  display.print("Current: "); display.print(curr); display.println("A");
  display.print("Status: "); display.println(br > 0 ? "ON" : "OFF");
  display.display();
}
```

BLOCK DIAGRAM

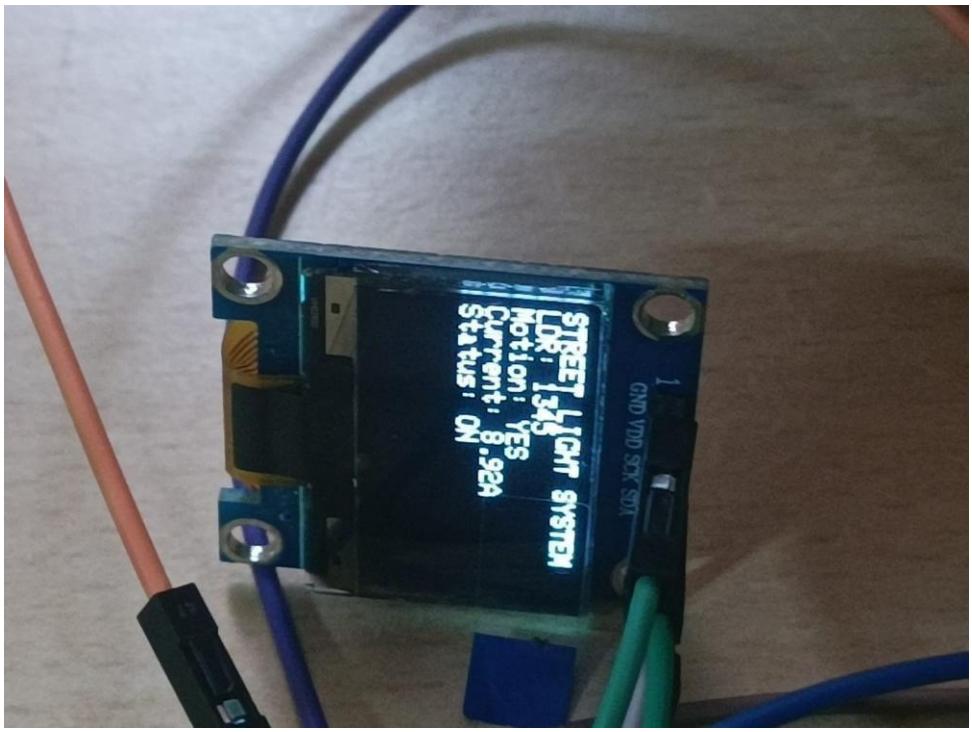


FLOW CHART


Start System
Initialize ESP32, Sensors, OLED, WiFi
Read LDR Sensor Value
Is it Dark?
Yes
No
Turn OFF LED Street Light
Read PIR Sensor
Motion Detected?
Yes
No
Keep Light OFF / Dim
Turn ON LED Street Light
Measure Current using Current Sensor
Display Data on OLED
Send Data to ThingSpeak Cloud

LOCAL OUTPUT

The OLED display provides real-time local feedback about the system status. It shows information such as sensor readings, LED status, and current consumption. This allows users to monitor system operation directly without requiring internet access.



CLOUD OUTPUT

The ThingSpeak dashboard displays real-time graphs representing sensor readings and system activity. The platform visualizes parameters such as ambient light level, motion detection status, current consumption, and LED operation.

RESULTS AND OBSERVATIONS

The implemented system performed successfully during testing. The LDR sensor accurately detected ambient light conditions and allowed the system to differentiate between day and night environments. The PIR motion sensor successfully detected movement and triggered the LED street light when motion was detected during dark conditions. The current sensor provided real-time energy consumption readings which were transmitted to the ThingSpeak platform. The cloud dashboard displayed graphs showing LDR values, motion detection status, LED activity, and energy usage. These results confirm that the system effectively demonstrates automated street lighting and energy monitoring.

INSIGHTS

- The Smart Street Lighting and Energy Optimization System demonstrates how IoT technologies can significantly improve energy efficiency in public infrastructure. Through the integration of sensors and cloud connectivity, the system is capable of automatically controlling street lights based on environmental conditions and motion detection. The LDR sensor enables the system to detect day and night conditions, while the PIR sensor detects human or vehicle movement to activate the street light only when required.
- The integration of the current sensor provides real-time information about energy consumption, allowing users to monitor power usage effectively. Additionally, the ThingSpeak cloud platform allows remote monitoring of system data through graphical visualization, which helps in analyzing energy consumption patterns and system performance over time.
- These insights highlight the potential of IoT-based smart lighting systems in reducing unnecessary energy usage and supporting smart city initiatives.

CHALLENGES

- During the development and testing of the system, several challenges were encountered. One of the primary challenges was sensor calibration. The LDR sensor readings can vary depending on environmental lighting conditions, making it necessary to carefully adjust threshold values for accurate day and night detection.
- Another challenge was ensuring reliable motion detection using the PIR sensor. The sensor may occasionally detect false triggers due to environmental factors such as temperature variations or sudden movements nearby.
- Network connectivity was also a challenge, as stable Wi-Fi connectivity is required for transmitting data to the ThingSpeak cloud platform. Any interruption in internet connectivity may temporarily affect data transmission.
- Additionally, integrating multiple sensors and ensuring proper communication between hardware components required careful configuration and testing.

PREVENTIVE RECOMMENDATIONS

- To improve the reliability and performance of the system, several preventive measures can be implemented. Proper calibration of the LDR sensor threshold values can ensure accurate detection of ambient light conditions. Additionally, placing the PIR sensor in an optimal position can help reduce false motion detection.
- Using stable and reliable Wi-Fi connectivity can improve the consistency of cloud data transmission. Implementing data buffering or local storage can also help prevent data loss during temporary network interruptions.
- For large-scale deployment, more robust lighting systems and industrial-grade sensors can be used to enhance system durability and accuracy. Implementing adaptive brightness control using PWM can further optimize energy usage by adjusting light intensity instead of simply turning lights on or off.

CONCLUSION

The Smart Street Lighting and Energy Optimization System successfully demonstrates how IoT technology can be used to improve energy efficiency in public lighting infrastructure. The system uses an LDR sensor to detect ambient light conditions and a PIR sensor to identify motion, allowing the street light to operate only when required. This automation helps reduce unnecessary energy consumption while maintaining safety in low-light environments. The ESP32 microcontroller processes sensor data and controls the LED street light accordingly. A current sensor monitors power consumption, while the OLED display provides real-time local information about system status. Additionally, sensor data is transmitted to the ThingSpeak cloud platform, enabling remote monitoring and graphical analysis of system performance. The implementation confirms that IoT-based smart lighting systems can effectively reduce energy wastage and support smart city initiatives. By integrating sensing, automation, and cloud monitoring, the system provides a practical and scalable solution for intelligent energy management in urban street lighting systems.